

Observation Model based Perceptually Motivated Bilateral Filter for Image Reconstruction

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Abstract—Recently, a lot of bilateral and non-local means based reconstruction algorithms are proposed in literature. The success of these filters lies in finding similar patches in the neighbourhood. However, in complex regions and in presence of noise, finding similar patches based on mean square error (MSE) is not reliable. This results into blurred edges and visible patches in the reconstructed image. To address this issue, we propose a new Observation model based and Perceptually Motivated Bilateral Filter (OPBIF) for Image Reconstruction. In which image quality assessment (IQA) matrices are used to find the similarity among the patches. From experimental results, it is validated that the proposed algorithm has the capability of reconstructing sharp edges, as compared to existing non-linear filtering algorithms.

Index Terms—Bilateral filter, IQA matrices, image reconstruction, perceptually motivated.

I. INTRODUCTION

Image reconstruction is an important phenomenon, which requires either the estimation of missing pixels (interpolation, deinterlacing, removal of random valued and fixed valued impulse noise, etc.) or the rectification of high frequency information of the received data (de-blocking of JPEG-2000/Jpeg, denoising etc.). Existing reconstruction algorithms can be broadly divided into three categories, 1) Geometric duality, 2) Sparse coding/ learning and 3) Non-linear (bilateral, non-local means or Kernel regression [10]) filtering based reconstruction. In this work, we are going to concentrate only on non-linear filtering based reconstruction [1-3], [10] .

The bilateral filter [1] was first proposed by Tomasi for removing Gaussian noise, in which neighboring pixels are assigned weights based upon the pixel intensity difference (radiometric distance). On the same line, in a non-local means filter [2] mean square error between patches is used instead of pixel intensity difference for assigning the weights. Inspired by the success of bilateral and non-local means filters, these filters are extensively studied [3] and used in several image processing and computer vision applications. For image interpolation purpose, Hung et. al [4] estimated prediction parameters using a bilateral filter instead of covariance matrix [5], [15] based methods. In [6], authors used bilateral filters to alleviate the effect of “geometric duality” mismatching issue in soft-decision adaptive interpolation (SAI) algorithm [5].

For deinterlacing Wang et. al [11] and Jakhetiya et. al [9] used a non-local means and a observation based bilateral filter

(OBF) respectively to predict missing pixels of a frame. These non-local filters are also extensively used for image denoising. Along the same lines, Xiong et. al [12] and Hsing et. al [13] used non-local means [2] and bilateral filters [1] respectively for removing mixed noise (both Gaussian and Random valued impulse noise). These algorithms produce satisfactory results when noise density is moderate. However, for high level of noise density, reconstructed image has visible patches.

Rehman et. al [16] made first attempt to use image quality assessment (IQA) matrix (SSIM) to restore the edges of non-local means [2] reconstructed image. Unfortunately, this work is only constrained to image denoising and does not come into the category of non-linear filtering.

From, applications given above, it can be observed that these filters [1-2] play a vital role in image reconstruction. As described earlier in these filters, mean square error between patches/pixels is used to estimate the similarity between patches and subsequently, weights are assigned depending upon the patch similarity. Unfortunately, in noisy situations mean square error (MSE) is not reliable, as MSE is more sensitive to intensity variations [7] than edge semantics. This may result in wrong edge direction estimation and/or blurred edges.

To overcome the above described problems associated with bilateral and non-local mean filters, we propose an observation model based [9] perceptually motivated bilateral filter for image reconstruction, in which gradient magnitude similarity deviation [8] (GSMD) based IQA matrix is used for estimating the similarity between patches. To validate our proposed algorithm, we compare observation model based perceptually motivated bilateral filter (OPBIF) with bilateral [1], non-local means [2] and kernel regression [10] for irregularly sampled data and with kernel regression [10], SAI [5], SME [17] and directional bicubic [18] for the interpolated image.

The rest of paper is organised as follows. Section 2 briefly describes the gradient magnitude similarity deviation [8] (GSMD) and the proposed OPBIF filter. The comparison and implementation results are provided in Section 3 and the concluding remarks are given in Section 4.



Fig. 1. Comparison of subjective quality for reconstructed Lena image whose 65% of the pixels were removed. (a) Irregularly sampled data, restored image with (b) EPA (c) Bilateral filter (d) Non-local means filter (e) Kernel Regression (f) Proposed Algorithm respectively.

II. PROPOSED OBSERVATION MODEL BASED PERCEPTUALLY MOTIVATED BILATERAL FILTER

Recently, an observation based bilateral filter (OBF) [9] was proposed for predicting missing lines of the interlaced frame. This filter has the capability of incorporating both the prior and the likelihood. The OBF [9] filter works well and produces better results than the ordinary bilateral filter when neighboring pixels are reliable (such as in the deinterlacing application) for the radiometric distance (pixel similarity) estimation.

However, an OBF filter performs poorly in situations where neighboring pixels are noisy (denoising, irregularly sampled data, etc.). In these situations, MSE is more sensitive to noise patterns than edge semantics and more weight is assigned to similar noisy pixels than those with similar edge semantics. So, ordinary and observation based bilateral filters both are not favorable in these applications and produce blurry edges and visible patches in the restored image, especially when level of noise density is high or neighboring pixels are not reliable.

A lot of image quality assessment (IQA) [7] algorithms

have been proposed, which are successfully able to predict the quality of the restored image which matches the visual perception of the human eye. With the success of IQA matrices [7], [8] we incorporated the recently developed gradient magnitude similarity deviation [8] (GSMD) based IQA matrix with the bilateral filter to address the problems associated with the ordinary and observation model based bilateral filters [9]. We call our proposed filter as Observation Model based and Perceptually Motivated Bilateral Filter (OPBIF).

In this paper, we apply the proposed OPBIF filter for reconstructing the missing pixels of the irregularly sampled data as image interpolation, deinterlacing and denoising (Random valued impulse noise and Fixed valued impulse noise) are the special cases of the irregularly sampled data. We formulate reconstruction of the missing pixels of irregularly sampled data problem as follows:

$$B^* = \arg \min_B \sum_{A_{ik} \in N_i} (w_{ik}(B_i - A_{ik})^2) + \lambda_i(B_i - \tilde{B}_i)^2 \quad (1)$$

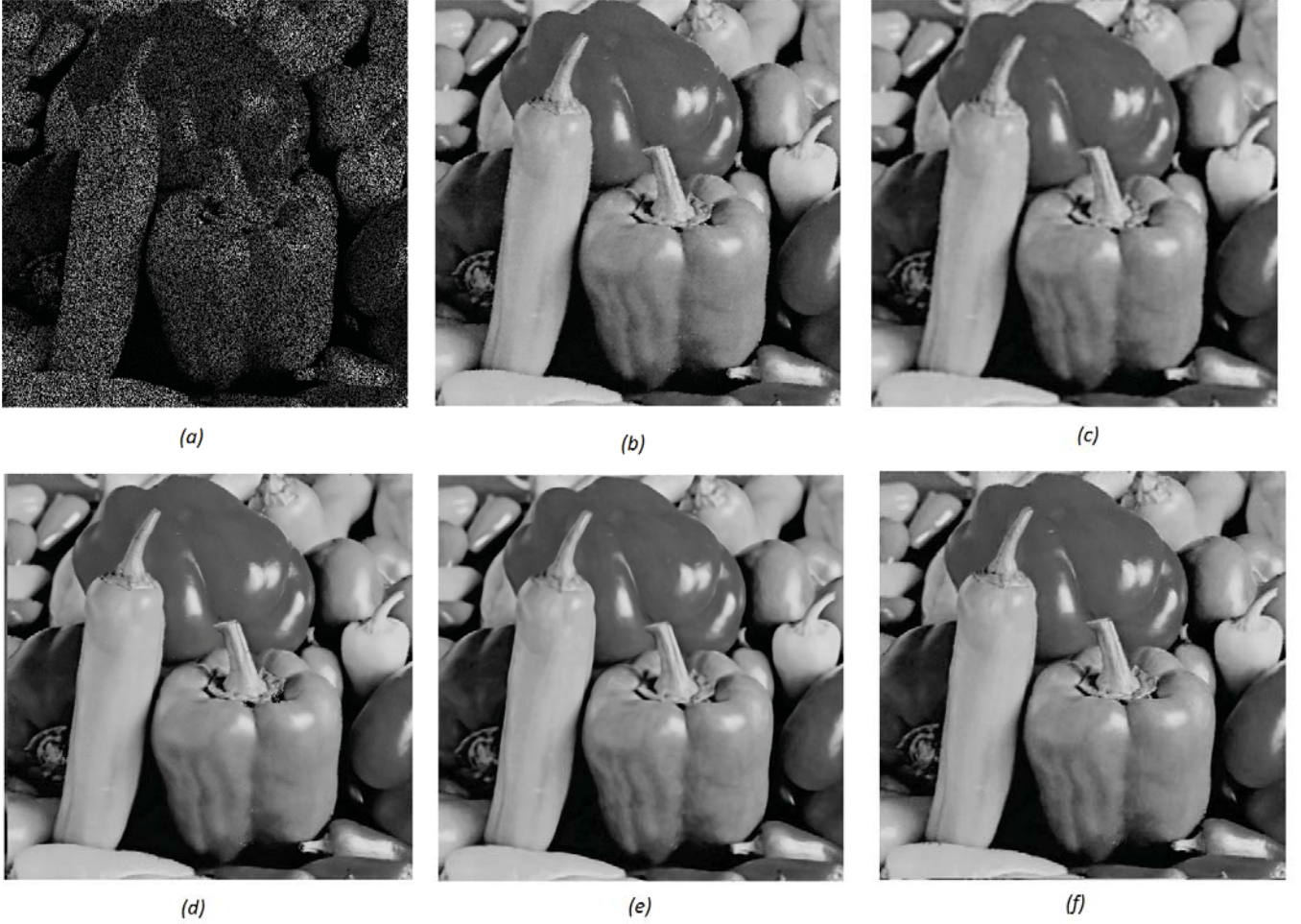


Fig. 2. Comparison of subjective quality for reconstructed Peppers image whose 65% of the pixels were removed. (a) Irregularly sampled data, restored image with (b) EPA (c) Bilateral filter (d) Non-local means filter (e) Kernel Regression (f) Proposed Algorithm respectively.

Here, B_i and $\tilde{B}_i$ are the missing pixels and approximation of the missing pixels respectively. These approximated pixels can be generated using any of the reconstruction algorithms. In proposed algorithm, we choose edge preserving algorithm (EPA) [14] for the reconstruction of the approximated pixels ($\tilde{B}_i$). The optimal solution of the observation model based bilateral filter (OBF) [9] in closed form is obtained by differentiating the (1) and equating to zero.

$$B_i^* = \frac{\lambda \tilde{B}_i + \sum w_k A_{ik}}{\lambda + \sum w_{ik}} \quad (2)$$

Here, A_{ik} are the neighboring pixels in 5×5 window (search range) centered at the missing pixel B_i used for the reconstruction. While λ is the controlling parameter, which provides trade-off between the prior and the likelihood. On the other hand, w_{ik} are the weighting parameters corresponding to the pixels in the window A_{ik} . The simplest way to estimate these weighting parameters is to use pixel difference as used in the ordinary and observation based bilateral filter.

$$w_{ik} = \exp\left(-\frac{\|\tilde{B}_i - A_{ik}\|_2}{2\sigma_1}\right) \exp\left(-\frac{\|D - D_k\|_2}{2\sigma_2}\right) \quad (3)$$

As we discussed earlier MSE or pixel difference is not able to perform efficiently, in the situation where neighboring pixels are not reliable in the searching window. We did extensive experiments and observed that IQA matrices have much better ability of finding similar patches in noisy situations than MSE and radiometric distance. This motivates us to incorporate an IQA matrix and a bilateral filter into our approach as:-

$$w_{ik} = \exp\left(-\frac{GMSD(\tilde{B}_i, A_{ik})}{2\sigma_1}\right) \exp\left(-\frac{\|D - D_k\|_2}{2\sigma_2}\right) \quad (4)$$

Here, second term in the proposed OPBIF is the pixel location distance between the pixel of interest B_i and neighboring pixel A_{ik} . GMSD [8] (gradient magnitude similarity deviation) is the recently developed IQA matrix GMSD [8] is based upon the variation of the edge strength. This variation of edge strength and edge strength in GMSD are estimated as follows:-

TABLE I
COMPARISON OF SIMULATION RESULTS IN TERMS OF PSNR (IN dB) AND
GMSD (IN %) FOR LENA IMAGE WHOSE 65% OF THE PIXELS ARE
REMOVED.

Image	[14]	[1]	[2]	[10]	OPBIF
PSNR	31.562	28.602	32.159	31.773	32.636
GMSD (in %)	4.173	7.726	4.171	4.879	3.983

TABLE II
COMPARISON OF SIMULATION RESULTS IN TERMS OF PSNR (IN dB) AND
GMSD (IN %) FOR PEPPERS IMAGE WHOSE 65% OF THE PIXELS ARE
REMOVED.

Image	[14]	[1]	[2]	[10]	OPBIF
PSNR	31.936	29.782	31.876	32.514	33.101
GMSD (in %)	3.513	6.915	3.479	4.152	3.150

$$GMSD = \sqrt{\frac{1}{N} \sum_{j=1}^N (GMS(j) - GMSM)^2} \quad (5)$$

$$GMS(j) = \frac{2m_{B_i}(j)m_{A_{ik}}(j) + c}{m_{B_i}^2(j) + m_{A_{ik}}^2(j) + c} \quad (6)$$

Here, $GMSM$ is the mean of the gradient magnitude similarity (GMS) of all pixels. While, m_{B_i} and $m_{A_{ik}}$ are the gradient magnitude of edge in horizontal and vertical direction respectively of the corresponding blocks.

III. EXPERIMENTAL RESULTS

The proposed observation model based perceptually motivated bilateral (OPBIF) filter is implemented and to show the efficiency of the proposed OPBIF filter, the proposed algorithm is compared with the ordinary bilateral filter [1], edge preserving filter [14], non-local means filter [2] and kernel regression [10] respectively, for reconstructing the missing pixels of the irregularly sampled image. It can be observed from Tables I-II that the proposed filter achieves higher PSNR and lower GMSD [8] as compared to other algorithms. A lower value of GMSD suggests that reconstructed image with the proposed filter is visually more appealing and produces sharp edges.

On the other hand, subjective comparison is shown in Figs. 1 and 2 for the reconstructed Lena and Pepper images with 65% of the pixels were removed respectively. Bilateral and non-local means filters produced blurry edges and visible patches in the noisy situations, near the complex regions, as shown in the Figs. 1 and 2 (near the hairs of reconstructed Lena image). While kernel regression [10] algorithm does not produce any visible artifacts but this algorithm suffers with the over-smoothed reconstructed image.

To show the applicability of the proposed OPBIF filter, we applied OPBIF filter also on the bicubic interpolated image and results are presented in the Table III. It can be observed that proposed filter achieves better results (in terms of $PSNR$) than the state of the art interpolation algorithms, such as Kernel regression [10], SAI [5], SME [17] and direction bicubic interpolation [18] algorithms. In this work, we had

TABLE III
COMPARISON OF RESTORATION RESULTS IN TERMS OF PSNR (dB) FOR
IMAGE INTERPOLATION.

Image	[10]	[5]	[17]	[18]	OPBIF
Tiffney	29.775	30.076	29.989	29.945	30.017
Baboon	22.973	23.234	23.180	23.051	23.391
Sar-Large	19.100	19.120	18.930	18.971	19.508
Aerial2	21.750	21.866	21.800	21.773	22.058
Ruler.512	6.101	11.217	11.170	11.687	12.212
Party	15.365	15.436	15.280	15.314	15.682
Bike	17.463	17.656	17.700	17.541	18.079
Average	14.725	19.801	19.721	19.755	20.135

shown simulation results only on the reconstruction of the irregularly sampled images and down-sampled images. While our initial results suggests that, the proposed algorithm has the capability of having “uniform framework” and can be applied to several other image processing reconstruction problems. We will investigate this issue in the near future.

IV. CONCLUSION

In this work, we had proposed a new observation model based perceptually motivated bilateral filter (OPBIF). The proposed OPBIF filter has the better capability of finding the similar patches in neighborhood as compared to ordinary bilateral filter and non-local means filters, even in the situation when neighboring pixels are noisy and unreliable. From simulation results, it is validated that proposed algorithm produced better reconstructed image as compared to existing non-linear filters, in terms of both objective and subjective quality.

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